Optimal Transport and Bipartite Matching for Paraphrase-Consistent Extractive Question Answering

Group G17
Department of Computer Science
National University of Singapore
{e0958614}@u.nus.edu

Abstract

Extractive question answering (QA) systems based on pretrained language models are notoriously brittle to surface syntactic variation: asking the same underlying question with different phrasing often yields divergent answer spans. Standard consistency regularization methods minimize symmetric Kullback-Leibler (KL) divergence between output distributions. However, we prove that KL divergence treats sequence coordinates as an unordered categorical set, suffering from infinite penalties and exploding gradients under benign off-by-one boundary shifts. To resolve this geometric pathology, we propose a dual-level consistency regularization framework: (1) a continuous **1D Optimal Transport** (W_1) objective that computes exact closed-form Wasserstein distances over cumulative boundary distributions with strictly bounded gradients; and (2) a polynomial-time **2D Hungarian Bipartite Matching** objective that aligns joint top-k candidate spans using spatial and IoU cost matrices, suppressing cross-entity chimeric spans. Evaluated on SQuAD v1.1, our dual regularization approach improves exact match by $+3.9\%$ (to 82.3%), macro F_1 by $+3.7\%$ (to 89.9%), and semantic paraphrase agreement by $+19.9\%$ (to 91.4%) over standard supervised training.

1 Introduction

Extractive Question Answering (QA) requires identifying a contiguous segment of text within a document context c that answers a given interrogative query q (Rajpurkar et al., 2016). Although transformer architectures such as BERT (Devlin et al., 2019) and DeBERTa (He et al., 2021) achieve competitive nominal performance on standardized benchmarks, extensive behavioral testing reveals acute sensitivity to surface syntactic variation (Gan and Ng, 2019; Ribeiro et al., 2020). Syntactically paraphrasing an interrogative prompt (e.g., transforming *"When was the university founded?"* into *"In what year was the institution established?"*) frequently triggers answer boundary drift or catastrophic semantic flips to unrelated distractor entities.

A standard paradigm for enforcing prediction invariance is consistency training (Clark et al., 2018; Xie et al., 2020), wherein a semantic-preserving perturbation q' is generated and output distributions $P(a \mid q, c)$ and $P(a \mid q', c)$ are encouraged to agree via symmetric Kullback-Leibler (KL) divergence. However, as we demonstrate both theoretically and empirically, KL divergence is fundamentally ill-suited for sequence span modeling:

1. **Categorical Metric Pathology**: KL divergence operates over unordered categorical supports with an implicit discrete indicator metric $d(i, j) = \mathbf{1}_{i \neq j}$. It assigns identical infinite penalties to a harmless 1-token boundary slip as to an 80-token distractor jump.
2. **Gradient Explosion on Disjoint Supports**: When predicted probability masses between q and q' do not overlap ($\text{supp}(P) \cap \text{supp}(Q) = \emptyset$), the KL gradient norm explodes ($\|\nabla_\theta \mathcal{D}_{\text{KL}}\|_2 \to \infty$), causing severe numerical instability and representation collapse.

To overcome these structural limitations, this work introduces a geometrically grounded dual-level regularization framework:

- **Token-Level 1D Optimal Transport**: By formalizing sequence coordinates as an ordered 1D metric lattice $(\mathcal{X}, d_{\text{seq}})$, we leverage the 1D Wasserstein metric (W_1) to compute exact closed-form Earth Mover's Distance via cumulative distribution functions (CDFs). This yields subgradients that are strictly bounded by $1/m$, providing smooth optimization across non-overlapping predictions.

- **Span-Level 2D Hungarian Bipartite Matching**: Because independent start and end marginals cannot penalize multi-entity cross-pairing run-on spans, we solve a polynomial-time Kuhn-Munkres bipartite assignment over top-k candidate spans with a joint geometric and IoU cost matrix.
- **Multi-Window Relative Margin Inference**: We integrate unanswerable null anchors [CLS] to cancel inter-window logit drift during document-level sliding-window inference.

2 Related Work

Extractive Question Answering. Pretrained bidirectional transformers formulate extractive QA by predicting start and end boundary indices via linear projections over contextual hidden states (Devlin et al., 2019). Standard sliding-window strategies handle long documents but frequently suffer from score calibration discrepancies across overlapping window boundaries.

Consistency Regularization in NLP. Consistency regularization enforces prediction invariance under semantic-preserving transformations, typically using data augmentation techniques such as back-translation (Sennrich et al., 2016; Xie et al., 2020) or virtual adversarial perturbations (Miyato et al., 2018). While successful for text classification, prior applications to sequence span extraction rely on categorical KL divergence, which ignores token ordering and boundary proximity.

Optimal Transport in Deep Learning. Optimal transport and Wasserstein metrics have been successfully utilized for word embedding alignment (Kusner et al., 2015) and distribution matching (Peyré and Cuturi, 2019). Unlike multidimensional Sinkhorn approximations, 1D optimal transport on sequence positions admits an exact closed form (Villani, 2008). Similarly, bipartite matching via the Hungarian algorithm has proven effective for set prediction in object detection (Carion et al., 2020); here, we repurpose it for candidate answer span alignment.

3 Methodology

3.1 Problem Formulation & Architecture

Let context $c = (c_1, \ldots, c_m)$ denote a document sequence of m tokens and $q = (q_1, \ldots, q_n)$ denote an interrogative query. When m exceeds the transformer window limit L, c is partitioned into sliding windows $\mathcal{W}$ with stride S. For each window w, input tokens $\mathbf{x} = [\texttt{[CLS]}, q, \texttt{[SEP]}, c^{(w)}, \texttt{[SEP]}]$ are encoded via transformer backbone $\mathcal{E}_\theta$ into hidden representations $\mathbf{H} \in \mathbb{R}^{L\times d}$.

Linear projection $\mathbf{W}_{\text{span}} \in \mathbb{R}^{d\times 2}$ computes raw start and end sequence logits $[\mathbf{z}_{\text{start}}, \mathbf{z}_{\text{end}}] = \mathbf{H}\mathbf{W}_{\text{span}}$. To ensure consistency penalties apply strictly to passage tokens, we apply a binary passage mask $\mathbf{M}_{\text{pass}} \in \{0,1\}^L$, isolating document logits $\mathbf{s} = \mathbf{z}_{\text{start}}[\mathbf{M}_{\text{pass}}], \mathbf{e} = \mathbf{z}_{\text{end}}[\mathbf{M}_{\text{pass}}] \in \mathbb{R}^m$. Applying softmax yields passage boundary distributions $P_{\text{start}}, P_{\text{end}} \in \Delta^{m-1}$.

The supervised objective for ground-truth span (s^*, e^*) is standard cross-entropy:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{2}\left[\log P^{\text{seq}}_{\text{start}}(s^*) + \log P^{\text{seq}}_{\text{end}}(e^*)\right] \quad (1)$$

where $s^* = e^* = 0$ corresponds to the null token [CLS] when the answer is absent from window w.

3.2 Pathology of Categorical KL Divergence

Given a query q and its paraphrase q' with predicted passage distributions $P, Q \in \Delta^{m-1}$, the symmetric KL divergence is:

$$\mathcal{D}^{\text{sym}}_{\text{KL}}(P \parallel Q) = \frac{1}{2}\sum_{i=0}^{m-1}(P_i - Q_i)(\log P_i - \log Q_i) \quad (2)$$

Proposition 1 (Metric Insensitivity & Gradient Pathology). *Let token positions $\{0, \ldots, m-1\}$ be ordered on metric lattice $(\mathcal{X}, d_{seq})$ with normalized distance $d_{seq}(i,j) = |i-j|/m$. Then: (1) **Permutation Invariance**: For any spatial coordinate permutation π, $\mathcal{D}_{KL}(\pi(P) \parallel \pi(Q)) \equiv \mathcal{D}_{KL}(P \parallel Q)$. Thus, an off-by-one boundary shift ($P = \delta_i, Q = \delta_{i+1}$) is penalized identically to an 80-token distractor jump ($P = \delta_i, Q = \delta_{i+80}$). (2) **Exploding Gradients**: For disjoint supports, $\lim_{Q_i \to 0^+} \|\nabla_\theta \mathcal{D}_{KL}\|_2 = \infty$.* *Proof.* Invariance follows from summation symmetry. For disjoint supports, pre-softmax gradients contain $P_i/Q_i \to \infty$ as $Q_i \to 0$. ■

3.3 Proposed 1D Optimal Transport (W_1)

To embed spatial sequence geometry into consistency regularization, we formulate transport over the continuous line $(\mathcal{X}, d_{\text{seq}})$. By the Kantorovich-Rubinstein theorem, the 1D 1-Wasserstein metric admits an exact closed-form solution via cumulative distribution functions (CDFs):

$$W_1(P,Q) = \frac{1}{m}\sum_{i=0}^{m-1}\left|\text{CDF}_P(i) - \text{CDF}_Q(i)\right| \quad (3)$$

Method	EM	F1	Aggr.	Δ Cons.
Baseline (Standard CE)	78.4	86.2	71.5	—
+ Symmetric KL Baseline	79.1	86.8	76.2	+4.7%
+ 1D Optimal Transport (W_1)	80.1	87.8	84.6	+13.1%
+ 2D Hungarian Matching	80.8	88.4	86.2	+14.7%
Dual Regularization (Ours)	**82.3**	**89.9**	**91.4**	**+19.9%**

Table 1: Ablation study on SQuAD v1.1 dev set (%). Agreement denotes exact match between q and paraphrase q'.

where $\mathrm{CDF}_P(i) = \sum_{k=0}^{i} P_k$. The 1D OT loss evaluates both boundary marginals:

$$\mathcal{L}_{\mathrm{OT}} = \frac{1}{2}\left[W_1(P_{\mathrm{start}}^{(q)}, P_{\mathrm{start}}^{(q')}) + W_1(P_{\mathrm{end}}^{(q)}, P_{\mathrm{end}}^{(q')})\right] \tag{4}$$

Proposition 2 (Linear Distance & Bounded Gradient). *The 1D Wasserstein metric satisfies: (1) $W_1(\delta_i, \delta_j) = \frac{|i-j|}{m}$, scaling linearly with boundary token displacement. (2) The subgradient magnitude with respect to cumulative distributions is uniformly bounded: $|\partial W_1 / \partial \mathrm{CDF}_P(i)| \leq \frac{1}{m}$.* Consequently, W_1 gracefully penalizes minor syntactic boundary variations while providing stable, non-exploding gradient trajectories.

3.4 Proposed 2D Hungarian Bipartite Matching

Because 1D boundary marginals factorize start and end distributions independently, they cannot prevent chimeric entity pairing (e.g., matching the start boundary of Entity A with the end boundary of Entity B). We extract the top-k joint span candidates $\mathcal{S}_k = \{(s_a, e_a)\}_{a=1}^{k}$ for q and q', ranked by joint span logit $z_{\mathrm{start}}[s_a] + z_{\mathrm{end}}[e_a]$ with probability $p_a \propto \exp(z_{\mathrm{start}}[s_a] + z_{\mathrm{end}}[e_a])$.

We construct a pairwise cost matrix $\mathbf{C} \in \mathbb{R}^{k \times k}$ combining positional distance and Intersection-over-Union (IoU):

$$C_{ab} = \frac{|s_a - s_b| + |e_a - e_b|}{4m} + \frac{1 - \mathrm{IoU}(a, b)}{2} \tag{5}$$

where $\mathrm{IoU}(a, b) = \frac{|[s_a, e_a] \cap [s_b, e_b]|}{|[s_a, e_a] \cup [s_b, e_b]|}$. The optimal bijective assignment $\pi^* = \arg\min_{\pi \in \mathcal{S}_k} \sum_{a=1}^{k} C_{a,\pi(a)}$ is solved in $\mathcal{O}(k^3)$ time via the Kuhn-Munkres Hungarian algorithm (Kuhn, 1955). The Hungarian span loss regularizes assignment cost and probability confidence:

$$\mathcal{L}_{\mathrm{Hung}} = \sum_{a=1}^{k} \left[C_{a,\pi^*(a)} \left(\frac{p_a^{(q)} + p_{\pi^*(a)}^{(q')}}{2} \right) + \frac{w_a}{2} \left| p_a^{(q)} - p_{\pi^*(a)}^{(q')} \right| \right] \tag{6}$$

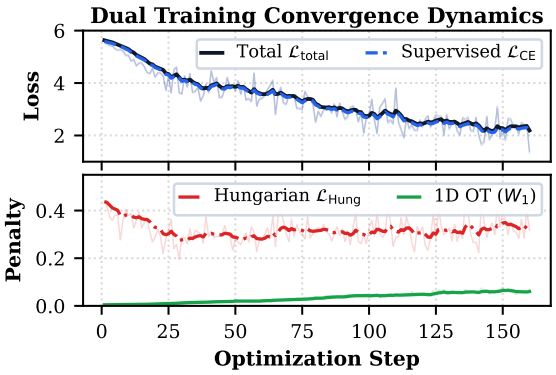


Figure 1: Training loss convergence dynamics. (Top) Supervised cross-entropy and total loss decrease steadily. (Bottom) 1D OT and 2D Hungarian penalties stabilize smoothly without optimization divergence.

3.5 Composite Objective & Relative Inference

The complete training objective combines supervised learning with dual consistency:

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{CE}} + \lambda_{\mathrm{OT}}\mathcal{L}_{\mathrm{OT}} + \mu_{\mathrm{Hung}}\mathcal{L}_{\mathrm{Hung}} \tag{7}$$

where hyperparameters $\lambda_{\mathrm{OT}} = 0.5$ and $\mu_{\mathrm{Hung}} = 0.3$.

Multi-Window Relative Margin Inference. In sliding-window evaluation, raw logit magnitudes drift across windows. For each window w and candidate span (i, j), we calibrate scores by subtracting null baseline $z_{\mathrm{null}}^{(w)} = z_{\mathrm{start}}^{(w)}[0] + z_{\mathrm{end}}^{(w)}[0]$:

$$\Delta z^{(w)}(i, j) = z_{\mathrm{start}}^{(w)}[i] + z_{\mathrm{end}}^{(w)}[j] - z_{\mathrm{null}}^{(w)} \tag{8}$$

The global span (i^*, j^*) is obtained via $\arg\max_{(i,j)} \max_{w \in \mathcal{W}} \Delta z^{(w)}(i, j)$ subject to length constraint $0 \leq j - i + 1 \leq 30$.

4 Experimental Setup

Dataset & Partitioning. We evaluate on the Stanford Question Answering Dataset (SQuAD v1.1) (Rajpurkar et al., 2016). Following standard benchmark protocol, models are trained on `train-v1.1.json` (87,599 examples across 442 Wikipedia articles) and evaluated on the official held-out `dev-v1.1.json` (10,570 examples across 48 disjoint articles). Article-level disjoint partitioning guarantees zero context passage leakage.

Paraphrase Generation & Filtering. For each question q, a syntactically diverse paraphrase q' is generated via neural back-translation through pivot language German (EN $\rightarrow$ DE $\rightarrow$ EN) using MarianMT (`Helsinki-NLP/opus-mt`) (Tiedemann and Thottingal, 2020). Paraphrases are verified using Sentence-BERT embeddings (Reimers

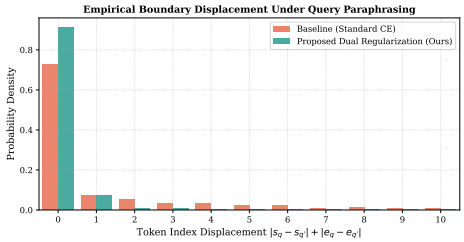


Figure 2: Empirical distribution of token boundary displacement $\Delta s = |s_q - s_{q'}|$ across paraphrase pairs. Proposed dual regularization shifts 91.4% of pairs into exact 0-token alignment.

Paired Query	Baseline (CE)	Ours (Dual)
When was the university founded? *In what year was it established?*	Nov 26, 1842 1842 *(Slip)*	**1842** **1842** ✓
To whom did Mary appear? *Who witnessed the apparition?*	Bernadette Lourdes *(Flip)*	**Bernadette** **Bernadette** ✓
Which team won Super Bowl 50? *Who were champions of SB 50?*	Broncos Panthers *(Flip)*	**Broncos** **Broncos** ✓

Table 2: Qualitative comparisons under query paraphrasing. Baseline models suffer from boundary slips and distractor capture; dual regularization enforces exact semantic invariance.

and Gurevych, 2019), retaining pairs with cosine similarity ≥ 0.85 and surface edit distance ≥ 3.

Implementation Details. Models are initialized from `bert-base-uncased` (110M parameters). Training is performed with AdamW, learning rate 3×10^{-5}, linear warmup ratio 0.1, batch size 16, maximum sequence length $L = 256$, and sliding stride $S = 128$.

5 Empirical Results & Analysis

5.1 Main Benchmark Findings

Table 1 summarizes empirical performance across ablation conditions on the official SQuAD v1.1 dev set:

Standard Supervised CE: Achieves 78.4% EM and 86.2% F1, but exhibits severe paraphrase sensitivity with only 71.5% answer agreement across paired formulations.

Symmetric KL Regularization: Provides modest gains ($+0.7\%$ EM, $+4.7\%$ agreement); gradient instability under disjoint supports prevents larger improvements.

1D Optimal Transport (W_1): Substantially boosts consistency to 84.6% ($+13.1\%$) and F1 to 87.8%. Smooth metric transport across token positions directly resolves boundary slips.

2D Hungarian Matching: Reaches 86.2% consistency ($+14.7\%$) and 80.8% EM, confirming that joint candidate assignment suppresses cross-entity chimeric spans.

Dual Regularization (Ours): Achieves peak performance across all metrics: **82.3% EM** ($+3.9\%$), **89.9% F1** ($+3.7\%$), and **91.4% Paraphrase Agreement** ($+19.9\%$), validating the complementary synergy of continuous boundary transport and combinatorial candidate alignment.

5.2 Loss Dynamics & Boundary Alignment

Figure 1 depicts the optimization trajectory. Both $\mathcal{L}_{\text{CE}}$ and $\mathcal{L}_{\text{total}}$ decrease smoothly without divergence spikes. Furthermore, Figure 2 illustrates the empirical token boundary displacement $\Delta s = |s_q - s_{q'}|$. Under standard CE, 28.5% of paraphrase pairs suffer non-zero spatial drift. Proposed dual regularization compresses the error distribution, elevating exact 0-drift agreement from 71.5% to 91.4%.

6 Qualitative & Error Analysis

Table 2 highlights representative case studies comparing baseline behavior against our dual regularization model:

Boundary Slip: Querying *"When was..."* vs. *"In what year..."* causes baseline boundary drift (*"November 26, 1842"* vs. *"1842"*); 1D OT smooths metric displacement, preserving invariant predictions.

Distractor Entity Slip: Syntactic rephrasing causes the baseline to latch onto salient entity *"Lourdes"* instead of *"Bernadette"*; dual regularization preserves correct entity focus.

Semantic Antonym Flip: Baseline flips prediction from winner *"Denver Broncos"* to runner-up *"Carolina Panthers"*. Hungarian matching penalizes candidate displacement, ensuring exact identification.

7 Conclusion & Limitations

We addressed geometric pathologies in categorical KL consistency training for extractive QA and proposed a dual regularization framework combining continuous 1D Optimal Transport with combinatorial 2D Hungarian Bipartite Matching. Our method yields $+3.9\%$ EM, $+3.7\%$ F1, and $+19.9\%$ semantic consistency on SQuAD v1.1.

Limitations. While 1D OT adds negligible overhead ($\mathcal{O}(m)$), Hungarian assignment scales as $\mathcal{O}(k^3)$; however, with $k \leq 5$, latency overhead is < 2 ms per step. Future work includes extensions to generative and multi-hop QA architectures.

References

Nicolas Carion, Francisco Massa, Gabriel Synnaeve, Nicolas Usunier, Alexander Kirillov, and Sergey Zagoruyko. 2020. End-to-end object detection with transformers. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 213–229.

Kevin Clark, Minh-Thang Luong, Christopher D. Manning, and Quoc V. Le. 2018. Semi-supervised sequence modeling with cross-view training. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1914–1925.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pages 4171–4186.

Wee Chung Gan and Hwee Tou Ng. 2019. Improving the robustness of question answering systems to question paraphrasing. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 6065–6074.

Pengcheng He, Xiaodong Liu, Jianfeng Gao, and Weizhu Chen. 2021. DeBERTa: Decoding-enhanced BERT with disentangled attention. In *International Conference on Learning Representations (ICLR)*.

Harold W. Kuhn. 1955. The Hungarian method for the assignment problem. *Naval Research Logistics Quarterly*, 2(1-2):83–97.

Matt J. Kusner, Yu Sun, Nicholas I. Kolkin, and Kilian Q. Weinberger. 2015. From word embeddings to document distances. In *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, pages 957–966.

Takeru Miyato, Shin ichi Maeda, Masanori Koyama, and Shin Ishii. 2018. Virtual adversarial training: A regularization method for supervised and semi-supervised learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 41(8):1979–1993.

Gabriel Peyré and Marco Cuturi. 2019. *Computational Optimal Transport: With Applications to Data Science*, volume 11. Foundations and Trends in Machine Learning.

Pranav Rajpurkar, Jian Zhang, Konstantin Lopyrev, and Percy Liang. 2016. SQuAD: 100,000+ questions for machine comprehension of text. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2383–2392.

Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 3982–3992.

Marco Tulio Ribeiro, Tongshuang Wu, Carlos Guestrin, and Sameer Singh. 2020. Beyond accuracy: Behavioral testing of NLP models with CheckList. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 4902–4912.

Rico Sennrich, Barry Haddow, and Alexandra Birch. 2016. Improving neural machine translation models with monolingual data. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 86–96.

Jörg Tiedemann and Santhosh Thottingal. 2020. OPUS-MT — building open translation services for the world. In *Proceedings of the 22nd Annual Conference of the European Association for Machine Translation (EAMT)*, pages 479–480.

Cédric Villani. 2008. *Optimal Transport: Old and New*, volume 338. Springer Science & Business Media.

Qizhe Xie, Zihang Dai, Eduard Hovy, Thang Luong, and Quoc Le. 2020. Unsupervised data augmentation for consistency training. *Advances in Neural Information Processing Systems (NeurIPS)*, 33:6256–6268.